Assignment # 2 MTH 501
Spring 2022
BC210209446
FOR WORD FILE:03066793776

Maximum Marks: 20
Due Date: 1 Sep, 2022

INSTRUCTIONS

Please read the following instructions before attempting the solution of this assignment:

- **To solve this assignment, you should have good command over 25-35 lessons.**
- **Try to get the concepts, consolidate your concepts and ideas from these questions which you learn in these lectures. You should concern the recommended books for clarification of concepts.**
- **Upload assignments properly through LMS. No Assignment will be accepted through email.**
- **Write your ID on the top of your solution file.**
- **Do not use colored backgrounds in your solution files.**
- **Use Math Type or Equation Editor etc. for mathematical symbols and equations.**
- **You should remember that if we found the solution files of some students are same then we will reward zero marks to all those students. Therefore, try to make solution by yourself and protect your work from other students, otherwise you and the student who send same solution file as you will be given zero marks.**
- **Avoid copying the solution from book (or internet); you must solve the assignment yourself.**
- **Also remember that you are supposed to submit your assignment in Word format any other like scan images, HTML etc. will not be accepted and we will give zero marks correspond to these assignments.**

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Question: 1 **Marks: 10**

Find the eigenvalues and eigenvectors of the following matrix?

$$A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$$

Solution:

First we find eigen values and eigen vectors

Eigen values:

$$(A - I\lambda) = 0$$

$$\left(\begin{bmatrix}1 & -1 & 4\\ 3 & 2 & -1\\ 2 & 1 & -1\end{bmatrix} - \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\lambda\right) = 0$$

$$\begin{bmatrix}1-\lambda & -1 & 4\\ 3 & 2-\lambda & -1\\ 2 & 1 & -1-\lambda\end{bmatrix} = 0$$

$$C_1 - (\lambda - 1)C_2 \quad and \quad C_3 + 4C_2$$

$$\begin{bmatrix}0 & -1 & 0\\ \lambda^2 - 3\lambda + 5 & 2-\lambda & 7-4\lambda\\ 3-\lambda & 1 & 3-\lambda\end{bmatrix} = 0$$

$$(0)(-1)^{1+1}\begin{vmatrix}2-\lambda & 7-4\lambda\\ 1 & 3-\lambda\end{vmatrix} + (-1)(-1)^{1+2}\begin{vmatrix}\lambda^2 - 3\lambda + 5 & 7-4\lambda\\ 3-\lambda & 3-\lambda\end{vmatrix} +$$

$$(0)(-1)^{1+3}\begin{vmatrix}\lambda^2 - 3\lambda + 5 & 2-\lambda\\ 3-\lambda & 1\end{vmatrix} = 0$$

$$(-1)(-1)^3\begin{vmatrix}\lambda^2 - 3\lambda + 5 & 7-4\lambda\\ 3-\lambda & 3-\lambda\end{vmatrix} = 0$$

$$\begin{vmatrix}\lambda^2 - 3\lambda + 5 & 7-4\lambda\\ 3-\lambda & 3-\lambda\end{vmatrix} = 0$$

$$(\lambda^2 - 3\lambda + 5).(3-\lambda) - (3-\lambda).(7-4\lambda) = 0$$

$$-\lambda^3 + 2\lambda^2 + 5\lambda - 6 = 0$$

$$-(\lambda - 3)(\lambda - 1)(\lambda + 2) = 0$$

$$(\lambda - 3)(\lambda - 1)(\lambda + 2) = 0$$

$$\lambda - 3 = 0 \quad , \quad \lambda - 1 = 0 \quad , \quad \lambda + 2 = 0$$

$$\lambda = 3 \quad , \quad \lambda = 1 \quad , \quad \lambda = -2$$

Now we find the Eigen Vectors

When $\lambda = 3$

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$$= \begin{bmatrix} 1-3 & -1 & 4 \\ 3 & 2-3 & -1 \\ 2 & 1 & -1-3 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -1 & 4 \\ 3 & -1 & -1 \\ 2 & 1 & -4 \end{bmatrix}$$

Apply row operation

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

To find the null space, solve the matrix equation

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

If we take $x_3 = t, then \;\; x_1 = t, \; x_2 = 2t$

Thus, $\vec{X} = \begin{bmatrix} t \\ 2t \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} t$

This is null space

Null space $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\}$

When $\lambda = 1$

$$= \begin{bmatrix} 1-1 & -1 & 4 \\ 3 & 2-1 & -1 \\ 2 & 1 & -1-1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 & 4 \\ 3 & 1 & -1 \\ 2 & 1 & -2 \end{bmatrix}$$

Apply row operation

$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \end{bmatrix}$$

To find the null space, solve the matrix equation

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$$\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & -4\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

If we take $x_3 = t, then \;\; x_1 = -t, \; x_2 = 4t$

Thus, $\vec{X} = \begin{bmatrix}-t\\ 4t\\ t\end{bmatrix} = \begin{bmatrix}-1\\ 4\\ 1\end{bmatrix} t$

The null space $\left\{\begin{bmatrix}-1\\ 4\\ 1\end{bmatrix}\right\}$

When $\lambda = -2$

$$= \begin{bmatrix}1-(-2) & -1 & 4\\ 3 & 2-(-2) & -1\\ 2 & 1 & -(-2)-1\end{bmatrix}$$

$$= \begin{bmatrix}3 & -1 & 4\\ 3 & 4 & -1\\ 2 & 1 & 1\end{bmatrix}$$

Apply row operation

$$= \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & -1\\ 0 & 0 & 0\end{bmatrix}$$

To find the null space, solve the matrix equation

$$\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & -1\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

If we take $x_3 = t, then \;\; x_1 = -t, \; x_2 = t$

Thus, $\vec{X} = \begin{bmatrix}-t\\ t\\ t\end{bmatrix} = \begin{bmatrix}-1\\ 1\\ 1\end{bmatrix} t$

The null space $\left\{\begin{bmatrix}-1\\ 1\\ 1\end{bmatrix}\right\}$

So, eigen value are

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3, 1, -2

Eigen vectors

$$\left\{\begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}-1\\4\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\1\end{bmatrix}\right\}$$

Question: 2 **Marks: 10**

Q # 2: Diagonalize the following matrix.

$$A = \begin{bmatrix}2 & 0 & 0\\1 & 2 & 1\\-1 & 0 & 1\end{bmatrix}$$

Solution:

First we find eigen values and eigen vectors

Eigen values:

$(A - I\lambda) = 0$

$$\left(\begin{bmatrix}2 & 0 & 0\\1 & 2 & 1\\-1 & 0 & 1\end{bmatrix} - \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}\lambda\right) = 0$$

$$\begin{bmatrix}2-\lambda & 0 & 0\\1 & 2-\lambda & 1\\-1 & 0 & 1-\lambda\end{bmatrix} = 0$$

$$(2-\lambda)(-1)^{1+1}\begin{vmatrix}2-\lambda & 1\\0 & 1-\lambda\end{vmatrix} + (0)(-1)^{1+2}\begin{vmatrix}1 & 1\\-1 & 1-\lambda\end{vmatrix} + (0)(-1)^{1+3}\begin{vmatrix}1 & 2-\lambda\\-1 & 0\end{vmatrix} = 0$$

$$(2-\lambda)(-1)^{2}\begin{vmatrix}2-\lambda & 1\\0 & 1-\lambda\end{vmatrix} = 0$$

$$(2-\lambda)(2-\lambda)(1-\lambda) = 0$$

$2-\lambda = 0 \quad , \quad 2-\lambda = 0 \quad , \quad 1-\lambda = 0$

$2 = \lambda \quad , \quad 2 = \lambda \quad , \quad 1 = \lambda$

$\Rightarrow \quad \lambda = 2 \quad , \quad \lambda = 2 \quad , \quad \lambda = 1$

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Now we find the Eigen Vectors

When $\lambda = 1$

$$= \begin{bmatrix} 2-1 & 0 & 0 \\ 1 & 2-1 & 1 \\ -1 & 0 & 1-1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & 0 \end{bmatrix}$$

Apply row operation

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \qquad R_2 - R_1 \ , \ R_3 + R_1$$

To find the null space, solve the matrix equation

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

If we take $x_3 = t, then \ \ x_1 = 0, \ x_2 = -t$

Thus, $\vec{X} = \begin{bmatrix} 0 \\ -t \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} t$

This is null space

Null space $\left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$

When $\lambda = 2$

$$= \begin{bmatrix} 2-2 & 0 & 0 \\ 1 & 2-2 & 1 \\ -1 & 0 & 1-2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ -1 & 0 & -1 \end{bmatrix}$$

Apply row operation

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$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad swap\ row\ 1\ and\ 2 \quad , \quad R_3 + R_1$$

To find the null space, solve the matrix equation

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

If we take $x_2 = t, then \;\; x_3 = s, \; x_1 = -s$

Thus, $\vec{X} = \begin{bmatrix} -s \\ -t \\ s \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} t + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} s$

The null space $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

So, eigen value 1, 2, 2
Eigen vectors

$$\left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Now, matrix P whose column entries are Eigen vectors

$$P = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

And matrix D whose column entries are Eigen Values

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

The matrices P and D are such that the initial matrix

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix} = PDP^{-1}$$

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